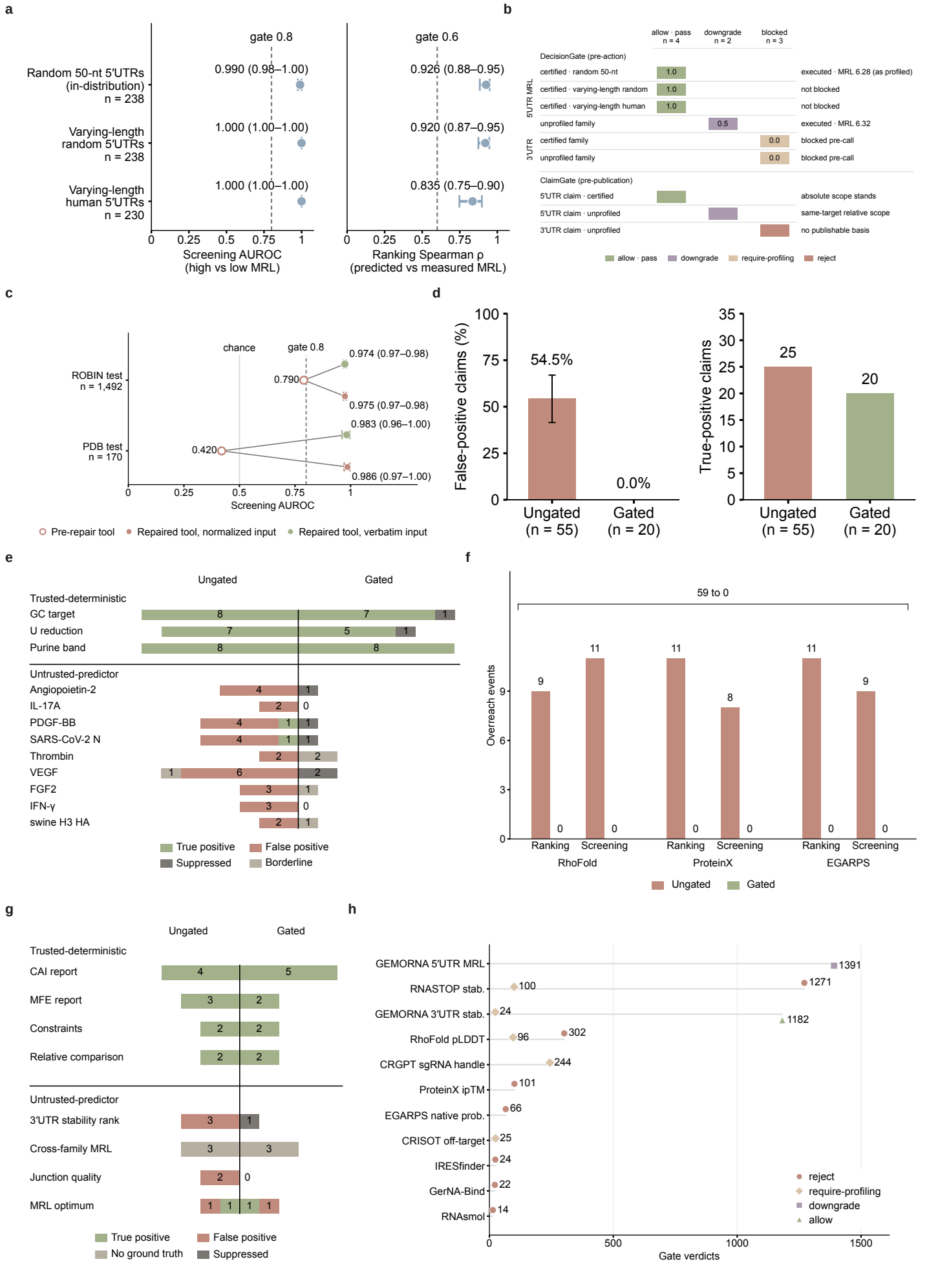

a
gate 0.8
gate 0.6
Random 50-nt 5′UTRs (in-distribution) n = 238
Varying-length random 5′UTRs n = 238
Varying-length human 5′UTRs n = 230
0.990 (0.98–1.00)
1.000 (1.00–1.00)
1.000 (1.00–1.00)
0.926 (0.88–0.95)
0.920 (0.87–0.95)
0.835 (0.75–0.90)
Screening AUROC (high vs low MRL)
Ranking Spearman ρ (predicted vs measured MRL)
b
allow · pass n = 4
downgrade n = 2
blocked n = 3
DecisionGate (pre-action)
5′UTR MRL
3′UTR
certified · random 50-nt
certified · varying-length random
certified · varying-length human
unprofiled family
certified family
unprofiled family
executed · MRL 6.28 (as profiled)
not blocked
not blocked
executed · MRL 6.32
blocked pre-call
blocked pre-call
ClaimGate (pre-publication)
5′UTR claim · certified
5′UTR claim · unprofiled
3′UTR claim · unprofiled
absolute scope stands
same-target relative scope
no publishable basis
allow · pass
downgrade
require-profiling
reject
c
chance
gate 0.8
ROBIN test n = 1,492
PDB test n = 170
0.790
0.420
0.974 (0.97–0.98)
0.975 (0.97–0.98)
0.983 (0.96–1.00)
0.986 (0.97–1.00)
Screening AUROC
Pre-repair tool
Repaired tool, normalized input
Repaired tool, verbatim input
d
False-positive claims (%)
54.5%
0.0%
Ungated (n = 55)
Gated (n = 20)
True-positive claims
25
20
e
Ungated
Gated
Trusted-deterministic
GC target
U reduction
Purine band
Untrusted-predictor
Angiopoietin-2
IL-17A
PDGF-BB
SARS-CoV-2 N
Thrombin
VEGF
FGF2
IFN-γ
swine H3 HA
True positive
False positive
Suppressed
Borderline
f
59 to 0
Overreach events
Ranking
Screening
RhoFold
ProteinX
EGARPS
Ungated
Gated
g
Ungated
Gated
Trusted-deterministic
CAI report
MFE report
Constraints
Relative comparison
Untrusted-predictor
3′UTR stability rank
Cross-family MRL
Junction quality
MRL optimum
True positive
False positive
No ground truth
Suppressed
h
GEMORNA 5′UTR MRL
RNASTOP stab.
GEMORNA 3′UTR stab.
RhoFold pLDDT
CRGPT sgRNA handle
ProteinX ipTM
EGARPS native prob.
CRISOT off-target
IRESfinder
GerNA-Bind
RNAsmol
reject
require-profiling
downgrade
allow
Gate verdicts